CLASS 9 — SCIENCE · FORMULA SHEET

Chapter 4: **Journey Inside the Atom** | Quick revision, one page

Sub-atomic particles

- Electron (J. J. Thomson): charge -1, mass $\approx \frac{1}{1836}$ u (negligible). Proton (Rutherford/Goldstein): charge $+1$, mass 1 u. Neutron (Chadwick, 1932): no charge, mass 1 u.
- Neutral atom: number of protons = number of electrons. Nucleus holds the protons and neutrons (nucleons).

Atomic number and mass number

- Atomic number Z = number of protons.
- Mass number A = protons + neutrons; number of neutrons $= A - Z$.
- Notation: ${}^{\mathrm{A}}_{\mathrm{Z}}\mathrm{X}$, e.g. ${}^{23}_{11}\mathrm{Na}$ has 11 protons, 11 electrons, 12 neutrons.

Electronic configuration

- Bohr's model: electrons revolve in fixed orbits (shells) K, L, M, N without radiating energy; energy changes only when an electron jumps between shells.
- Maximum electrons in shell n: $2n^2$ ($K = 2$, $L = 8$, $M = 18$); the outermost shell holds at most 8 electrons (2 for K).
- Examples: Li 2, 1; N 2, 5; Na 2, 8, 1; Ca 2, 8, 8, 2.

Valency

- Valency = number of electrons an atom gains, loses or shares to complete its outer shell (octet or duet).
- Valence electrons ≤ 4: valency = number of valence electrons (Na 1, Mg 2, Al 3). More than 4: valency $= 8-$ valence electrons (N 3, O 2, F 1).
- Noble gases (He, Ne, Ar) have complete outer shells and valency 0.

Isotopes and isobars

- Isotopes: same Z, different A (e.g. ${}^{35}\mathrm{Cl}$ and ${}^{37}\mathrm{Cl}$, protium, deuterium, tritium).
- Isobars: same A, different Z (e.g. ${}^{40}\mathrm{Ca}$ and ${}^{40}\mathrm{Ar}$).
- Average atomic mass $= \sum(\text{mass} \times \text{fraction of abundance})$; for Cl: $0.75 \times 35 + 0.25 \times 37 = 35.5$ u.